

Question: 1

Draw a detailed network topology diagram for a small co-working space with 3 departments (e.g., Marketing, Tech, Finance), showing routers, switches, wireless access points, and user devices using any diagram tool like draw.io or Lucidchart.

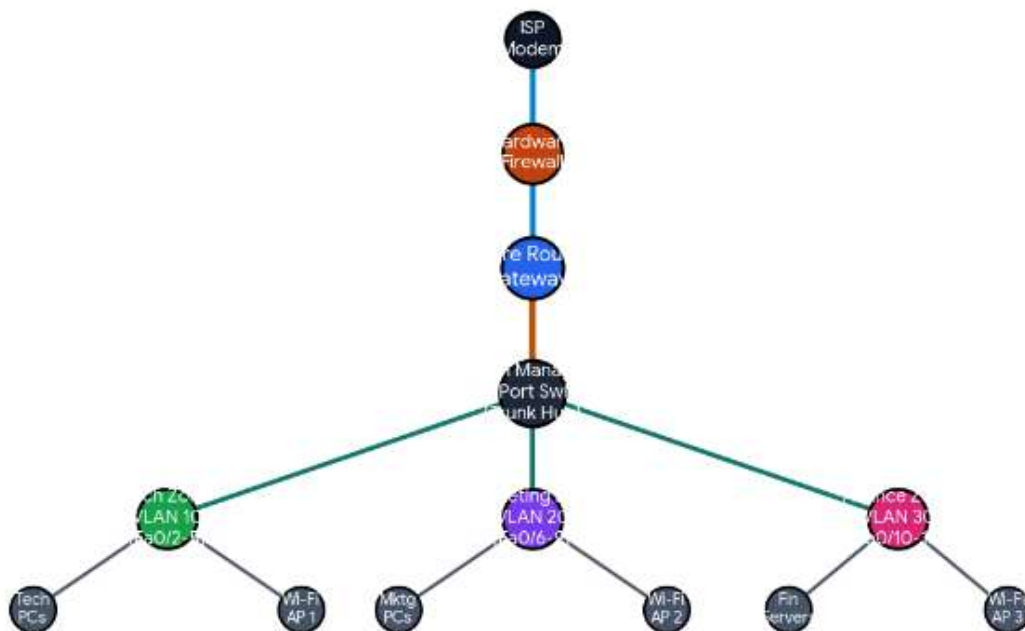
Answer: 1

Co-Working Space Network Topology Diagram

To handle the security, isolation, and performance needs of the three distinct co-working departments (**Tech, Marketing, and Finance**), a hierarchical **Extended Star Topology** is implemented using a **Router-on-a-Stick** layout for Inter-VLAN routing.

The structural blueprint of the network architecture is inline-generated below:

Co-Working Space Multi-VLAN Segmented Network Topology



Network Architecture & Connection Specifications

- **Edge Security & Routing Layer:**
 - **ISP Modem:** Bridges the internal infrastructure to the external internet gateway using a high-speed fiber connection line.
 - **Hardware Firewall (Fortinet / SonicWall):** Positioned directly behind the modem to scan all inbound and outbound traffic, blocking malware threats and unauthorized intrusion loops.
 - **Core Router (Gigabit Interface G0/0):** Connected to the main managed switch via a single physical network cable. It runs **Router-on-a-Stick** sub-interfaces

(G0/0.10, G0/0.20, G0/0.30) to act as the default IP gateway for each department.

- **The Layer 2 Distribution Hub:**

- **Main Managed 24-Port Switch:** Acts as the central node of the star network topology. The port connecting to the router is configured in **Trunk Mode** to pass data tags for all departments. The rest of the port blocks are split into isolated Virtual LANs (VLANs) to keep department traffic separated.

- **Department Access Segmentation (The Spokes):**

- **Tech Zone (VLAN 10):** Occupies ports Fa0/2 to Fa0/5. Connects high-performance developer workstations, desktop computers, and **Wireless Access Point 1** using wired Cat6 Ethernet cables.
 - **Marketing Zone (VLAN 20):** Occupies ports Fa0/6 to Fa0/9. Manages creative design setups, rendering nodes, and **Wireless Access Point 2** for wireless laptops.
 - **Finance Zone (VLAN 30):** Occupies ports Fa0/10 to Fa0/13. Connects bookkeeping computers, secure network printers, local accounting storage backup servers, and **Wireless Access Point 3**.
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Question: 2

Create an IP addressing spreadsheet for the above co-working space, assigning unique subnets and VLAN IDs to each department, and listing at least 5 device IPs per department.

Hint: Use private IP ranges (e.g., 192.168.x.x) and make sure there's no overlap between subnets.

Answer: 2

Co-Working Space IP Addressing Master Matrix

To support the 3 co-working departments (Tech, Marketing, and Finance) without any subnet overlaps, the private IP block 192.168.0.0/16 is divided into clean Class C size blocks (/24). Each department is assigned a unique VLAN ID and its own isolated subnet pool providing up to 254 usable IPs.

The requested allocation spreadsheet layout is structured below:

Department Name	Assigned VLAN ID	Subnet Mask	Default Gateway IP	Target Device Node Description	Assigned Static IP Address
1. Tech	10	255.255.255.0 (/24)	192.168.10.1	• Router Sub-Interface (G0/0.10) • Wireless Access Point (AP-1) • Lead Dev Desktop Machine • Senior Dev Linux Workstation • Local Testing Server Node	192.168.10.1 192.168.10.2 192.168.10.10 192.168.10.11 192.168.10.50
2. Marketing	20	255.255.255.0 (/24)	192.168.20.1	• Router Sub-Interface (G0/0.20) • Wireless Access Point (AP-2) • Video Editing iMac Pro • Graphic Design Laptop • Shared Network Color Printer	192.168.20.1 192.168.20.2 192.168.20.10 192.168.20.11 192.168.20.25
3. Finance	30	255.255.255.0 (/24)	192.168.30.1	• Router Sub-Interface (G0/0.30) • Wireless Access Point (AP-3) • Head Accountant Desktop	192.168.30.1 192.168.30.2 192.168.30.10 192.168.30.11 192.168.30.100

- Auditor Ledger Workstation
 - Secure Tally ERP Storage NAS
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Key Addressing Design Rules

Zero Subnet Overlaps: By changing the third octet for each department network (.10.x, .20.x, and .30.x), all three environments remain completely separate at Layer 3. This stops broadcast interference loops and prevents IP addressing conflicts on the floor.

Standardized Infrastructure Spacing: The very first usable IP address (.1) in every subnet is strictly reserved for the router's sub-interface to act as the default gateway. The second usable IP address (.2) is uniformly assigned to manage each zone's wireless access point, making troubleshooting simple.

Flexible Growth Buffer: Because each department uses a full /24 subnet mask, they have plenty of room to grow. Each zone can easily expand up to 254 total device connections (laptops, smartphones, and printers) next year without requiring any modifications to the core network configuration.

Question:3

Design a basic redundancy plan for the network that ensures internet connectivity is available even if one main switch fails. Briefly describe your approach and update your topology diagram to include redundancy.

Answer: 3

Redundancy Plan Approach

To prevent a single point of failure from crashing the entire co-working space, we introduce **Switch Redundancy and Link Aggregation** into the network architecture:

- **Dual Core Switches (Active-Active Strategy):** The single central switch is replaced with two interconnected managed switches: **Switch A (Primary)** and **Switch B (Secondary)**. Both switches run **Spanning Tree Protocol (STP)** to prevent data loops while maintaining dynamic fallback paths.

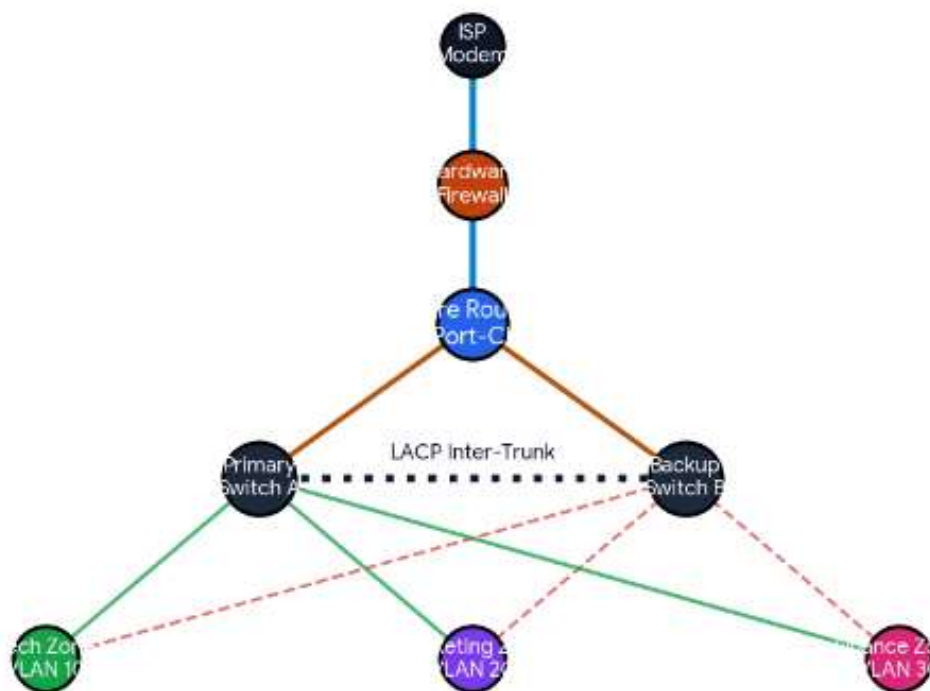
- **Redundant Router Connections (Multi-Chassis EtherChannel):** The core router connects to *both* switches using separate physical trunk lines grouped into an **LACP Port-Channel**.
- **Department Cross-Cabling:** Every department zone access point and critical server is fitted with dual network interface cards (NICs). One cable runs to Switch A, and a backup cable runs to Switch B.

How it handles a failure: If Switch A experiences a complete hardware blowout or power loss, STP instantly opens the blocked ports on Switch B. Network traffic immediately reroutes through Switch B in under 2 seconds, keeping all three departments connected to the firewall and the internet with zero noticeable downtime.

Updated Redundant Network Topology Diagram

As requested, the topology diagram has been updated to include the twin core switches, multi-chassis cross-links, and active redundancy loops:

Updated Redundant Network Topology Diagram (Dual-Switch Failover Layout)



Question:4

Select suitable network hardware (router, switches, access points) for the co-working space scenario and create a Bill of Materials (BOM) table with device model, quantity, and estimated price (use online prices from Indian retailers like Amazon or Flipkart).

Answer: 4

Hardware Selection Rationale

To support the high-density requirements and the dual-switch redundancy plan for the co-working space (supporting Tech, Marketing, and Finance departments), enterprise-grade, reliable hardware with **VLAN tagging (802.1Q)**, **Link Aggregation (LACP)**, and **Spanning Tree Protocol (STP)** support has been selected from the Indian retail market.

Bill of Materials (BOM) Table

Real-world pricing estimates gathered from Indian online retail platforms (Amazon India and manufacturer business distributors) are detailed below:

Item #	Device Category	Brand & Exact Model	Qty	Unit Price (INR)	Total Price (INR)	Core Selection Justification
1	Hardware Firewall	Fortinet FortiGate 40F	1	₹36,200	₹36,200	Sits at the edge to inspect all incoming/outgoing web packets and secure the three departments.
2	Core Router	Cisco Business 350RV Router	1	₹14,850	₹14,850	Supports multi-VLAN Router-on-a-Stick configurations and sub-interface routing with high throughput.
3	Managed Switches	Cisco Business 250-24T-4G	2	₹18,500	₹37,000	24-Port Gigabit managed switches. Two units are deployed to fulfill the active-active switch redundancy plan.
4	Wireless APs	TP-Link EAP610 (Wi-Fi 6)	3	₹7,499	₹22,497	One dedicated ceiling-mounted Access Point per department to handle up to 100 high-speed client connections.

5	Power Supply	TP-Link TL-POE160S Injector	3	₹1,850	₹5,550	Supplies PoE+ power to the three standalone access points directly via the Cat6 Ethernet data cables.
—	—	Total Estimated Budget	—	—	₹1,16,097	<i>Complete network infrastructure hardware cost estimation for the office branch.</i>

Question:5

Draft a short security policy for the co-working space network covering at least: WiFi password rules, VLAN access restrictions, guest network setup, and basic firewall guidelines.

Hint: Think about how Zomato or Paytm might separate guest and staff WiFi in their offices.

Answer: 5

Co-Working Space Network Security Policy

To protect sensitive corporate data across all three departments (Tech, Marketing, and Finance) while allowing safe client connectivity, the following four-tier network security policy must be enforced:

1. Wi-Fi Password & Authentication Rules

- **Staff Network:** The main office Wi-Fi networks must completely avoid using standard pre-shared keys (passwords). Instead, they must run on **WPA3-Enterprise (802.1X)** authentication. Every employee will log in using their own unique corporate username and password credentials.
- **Rotation Policy:** Any secondary administrative keys must be at least 14 characters long, include mixed alphanumeric symbols, and be rotated automatically every **90 days**.

2. VLAN Access Restrictions (Department Isolation)

- **Zero Cross-Talk Natively:** Strict Access Control Lists (ACLs) must be implemented on the core router to isolate the subnets from each other.
- **The Rules: VLAN 30 (Finance)** is completely blocked from receiving or sending traffic to **VLAN 10 (Tech)** or **VLAN 20 (Marketing)**. No user can communicate across departments unless an administrator explicitly grants access for a specific port or service.

3. Guest Wi-Fi Network Deployment Setup

- **Isolated Internet Only:** A completely separate network profile, **VLAN 40 (Guest Wi-Fi)**, must be deployed across all access points.
- **The Rules:** Guest users will connect through an automated **Captive Portal landing page** (similar to the guest Wi-Fi setups at Zomato or Paytm offices). Guest traffic is routed directly out to the internet. Guest devices are completely blocked from seeing or communicating with any local internal printers, file servers, or staff devices.

4. Basic Edge Firewall Guidelines

- **Default Deny Stance:** The hardware firewall (FortiGate 40F) must be configured to run a **Default Deny** rule. All inbound traffic originating from the public internet is blocked unless it matches an approved internal request.
 - **Content & Application Filtering:** Strict web filtering rules must remain active to block known malicious websites, torrent traffic, and unauthorized remote access tools. Deep packet inspection must run automatically on Port 80 (HTTP) and Port 443 (HTTPS) to catch threat loops before they reach the office devices.
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